Electric car: launch the design

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Plan a safe team workflow and identify the car's energy pathway.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 16.*
No. Faraday's law requires a change in flux, not merely nonzero flux.

02. Worked example: explain the model
3 m distance with clear timing boundaries; repeatable release and surface; speed distance/time.

03. Read and interpret the model
Accept a feasible labeled sketch with clear moving-part clearance.

04. Apply the physics
Roles may rotate; teacher authorizes tool use and checks setup, secured work, appropriate eye protection and clear power-test area.

05. Explain your reasoning
Example: complete the measured course within a chosen time while staying in lane; distance, time and lane observations provide evidence.

06. Review: connect an earlier idea *Review: Unit 7, Day 7.*
20 V; drops add to the supply voltage.

07. Exit check
Motor; teacher-approved wiring, secure parts, clear wheels and safe area.

08. STRETCH: test the model
Example finish within a marked lane on three runs; measure time and lateral deviation, then compare tradeoffs.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 16.*
Grid effects depend on geometry, grounding, resistance, controls and spatially varying fields. A single loop does not represent the whole network.

02. Worked example: explain the model
3 m distance with clear timing boundaries; repeatable release and surface; speed distance/time.

03. Read and interpret the model
Accept a feasible labeled sketch with clear moving-part clearance.

04. Apply the physics
1.2 W input; motor heating, drivetrain friction, rolling losses, wheel rotation, sound and drag.

05. Explain your reasoning
Example: complete the measured course within a chosen time while staying in lane; distance, time and lane observations provide evidence.

06. Review: connect an earlier idea *Review: Unit 7, Day 7.*
New resistance (4+2) $R=6\ R$; current ratio $4/6 \approx 0.6667$.

07. Exit check
Measured course, consistent start/stop, repeated runs, matched battery/surface/setup.

08. STRETCH: test the model
Specific measurable targets and all-trial reporting; a prespecified mean/median and completion fraction avoid cherry-picking.